

Linear algebra: Exercise chapter 9

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Autumn 2025

1. For the following subspaces S of $\mathbb{R}^n$, find the orthogonal complement $S^\perp$:

$$S_1 = \text{span} \left(\begin{bmatrix} 0 \\ 0 \\ 0 \end{bmatrix} \right), \quad S_2 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \right), \quad S_3 = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \\ -1 \end{bmatrix} \right), \quad S_4 = \text{span} \left(\begin{bmatrix} 1 \\ 2 \\ 2 \\ 3 \end{bmatrix}, \begin{bmatrix} 1 \\ 3 \\ 3 \\ 2 \end{bmatrix} \right).$$

Answer.

Subspace S_1 :

$$S_1^\perp = \{x \in \mathbb{R}^3 : \langle x, u \rangle = 0, \forall u \in S_1\} = \{x \in \mathbb{R}^3 : \langle x, 0 \rangle = 0\} = \mathbb{R}^3.$$

Subspace S_2 :

$$S_2^\perp = \{x \in \mathbb{R}^3 : \langle x, u \rangle = 0, \forall u \in S_2\} = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0\} = \text{span} \left(\begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix} \right).$$

Subspace S_3 :

$$\begin{aligned} S_3^\perp &= \{x \in \mathbb{R}^3 : \langle x, u \rangle = 0, \forall u \in S_3\} = \{x \in \mathbb{R}^3 : x_1 + x_2 + x_3 = 0, \quad x_1 + x_2 - x_3 = 0\} \\ &= \{x \in \mathbb{R}^3 : x_1 = -x_2, \quad x_3 = 0\} = \text{span} \left(\begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right). \end{aligned}$$

Subspace S_4 :

$$\begin{aligned} S_4^\perp &= \{x \in \mathbb{R}^4 : \langle x, u \rangle = 0, \forall u \in S_4\} = \{x \in \mathbb{R}^4 : x_1 + 2x_2 + 2x_3 + 3x_4 = 0, \quad x_1 + 3x_2 + 3x_3 + 2x_4 = 0\} \\ &= \{x \in \mathbb{R}^4 : x_4 = x_2 + x_3, \quad x_1 = -5x_2 - 5x_3\} = \text{span} \left(\begin{bmatrix} -5 \\ 1 \\ 0 \\ 1 \end{bmatrix}, \begin{bmatrix} -5 \\ 0 \\ 1 \\ 1 \end{bmatrix} \right). \end{aligned}$$

2. True or false:

- (a) The subspace spanned by $(1, 1, 0, 0, 0)$ and $(0, 0, 0, 1, 1)$ is the orthogonal complement of the subspace spanned by $(1, -1, 0, 0, 0)$ and $(2, -2, 3, 4, -4)$.
- (b) Two subspaces that only meet in the zero vector are orthogonal.

Answer.

(a): False. Define $u_1 := (1, 1, 0, 0, 0)$, $u_2 := (0, 0, 0, 1, 1)$, $v_1 := (1, -1, 0, 0, 0)$, $v_2 := (2, -2, 3, 4, -4)$, $S_1 = \text{span}(u_1, u_2)$, and $S_2 = \text{span}(v_1, v_2)$. It is evident that $\langle u_1, v_1 \rangle = 0$, $\langle u_1, v_2 \rangle = 0$, $\langle u_2, v_1 \rangle = 0$, and $\langle u_2, v_2 \rangle = 0$. Thus, S_1 and S_2 are orthogonal and $\dim(S_1) + \dim(S_2) = 2 + 2 = 4 \neq 5$. Hence, S_1 is not the orthogonal complement of S_2 .

(a): False. The subspaces $\text{span} \left(\begin{bmatrix} 1 \\ 0 \end{bmatrix} \right)$ and $\text{span} \left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} \right)$ in $\mathbb{R}^2$ meet only at the zero vector, but they are not orthogonal, $\left\langle \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 1 \\ 1 \end{bmatrix} \right\rangle = 1$.

3. Project the vector b onto the line through a :

$$(a) \quad b = \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix}, \quad a = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}.$$

$$(b) \ b = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}, \ a = \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix}.$$

Compute the projection matrices for the previous exercise and use these to verify your result.

Answer.

(a):

$$p = \frac{\langle b, a \rangle}{\langle a, a \rangle} a = \frac{5}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 5/3 \\ 5/3 \end{bmatrix}.$$

$$P = \frac{aa^T}{\langle a, a \rangle} = \frac{1}{3} \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 1 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix}, \quad Pb = \frac{1}{3} \begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 2 \\ 2 \end{bmatrix} = \frac{1}{3} \begin{bmatrix} 5 \\ 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 5/3 \\ 5/3 \\ 5/3 \end{bmatrix}.$$

(b):

$$p = \frac{\langle b, a \rangle}{\langle a, a \rangle} a = \frac{-11}{11} \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

$$P = \frac{aa^T}{\langle a, a \rangle} = \frac{1}{11} \begin{bmatrix} -1 \\ -3 \\ -1 \end{bmatrix} \begin{bmatrix} -1 & -3 & -1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 1 & 3 & 1 \\ 3 & 9 & 3 \\ 1 & 3 & 1 \end{bmatrix}, \quad Pb = \frac{1}{11} \begin{bmatrix} 1 & 3 & 1 \\ 3 & 9 & 3 \\ 1 & 3 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix} = \frac{1}{11} \begin{bmatrix} 11 \\ 33 \\ 11 \end{bmatrix} = \begin{bmatrix} 1 \\ 3 \\ 1 \end{bmatrix}.$$

4. Project b onto the column space of A . Make use of the projection matrix.

$$(a) \ b = \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix}, \ A = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix}.$$

$$(b) \ b = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}, \ A = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix}.$$

Answer.

(a):

$$A^T A = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 1 \\ 1 & 2 \end{bmatrix}, \quad (A^T A)^{-1} = \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix}.$$

$$P = A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 1 \\ 0 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 2 & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 \\ -1 & 1 \\ 0 & 0 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}.$$

$$p = Pb = \begin{bmatrix} 1 & 0 & 0 \\ 1 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} 2 \\ 3 \\ 4 \end{bmatrix} = \begin{bmatrix} 2 \\ 5 \\ 0 \end{bmatrix}$$

(b):

$$A^T A = \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} = \begin{bmatrix} 2 & 2 \\ 2 & 3 \end{bmatrix}, \quad (A^T A)^{-1} = \frac{1}{2} \begin{bmatrix} 3 & -2 \\ -2 & 2 \end{bmatrix}.$$

$$P = A(A^T A)^{-1} A^T = \begin{bmatrix} 1 & 1 \\ 1 & 1 \\ 0 & 1 \end{bmatrix} \begin{bmatrix} \frac{3}{2} & -1 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & 0 \\ \frac{1}{2} & 0 \\ -1 & 1 \end{bmatrix} \begin{bmatrix} 1 & 1 & 0 \\ 1 & 1 & 1 \end{bmatrix} = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix}.$$

$$p = Pb = \begin{bmatrix} \frac{1}{2} & \frac{1}{2} & 0 \\ \frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix} = \begin{bmatrix} 4 \\ 4 \\ 6 \end{bmatrix}$$

5. Find an orthonormal basis for the space spanned by:

$$a = \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \ b = \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \ c = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix}.$$

Answer.

$$v_1 = a, \quad v_2 = b - \frac{\langle v_1, b \rangle}{\langle v_1, v_1 \rangle} v_1 = b - \frac{0}{6} v_1 = b, \quad v_3 = c - \frac{\langle v_1, c \rangle}{\langle v_1, v_1 \rangle} v_1 - \frac{\langle v_2, c \rangle}{\langle v_2, v_2 \rangle} v_2 = \begin{bmatrix} 1 \\ 0 \\ 4 \end{bmatrix} - \frac{9}{6} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix} - \frac{1}{2} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix} = \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$
$$q_1 = \frac{1}{\sqrt{6}} \begin{bmatrix} 1 \\ 1 \\ 2 \end{bmatrix}, \quad q_2 = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ -1 \\ 0 \end{bmatrix}, \quad q_3 = \frac{1}{\sqrt{3}} \begin{bmatrix} -1 \\ -1 \\ 1 \end{bmatrix}.$$

6. Consider the matrix

$$A = \begin{bmatrix} 1 & 0 & 3 \\ 0 & -1 & 2 \end{bmatrix}.$$

Find orthonormal bases for the following spaces:

- Column space $C(A)$
- Row space $C(A^T)$
- Nullspace $N(A)$
- Left nullspace $N(A^T)$

Answer.

Column space $C(A)$

The columns of A are

$$c_1 = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad c_2 = \begin{bmatrix} 0 \\ -1 \end{bmatrix}, \quad c_3 = \begin{bmatrix} 3 \\ 2 \end{bmatrix}.$$

Since $c_3 = 3c_1 - 2c_2$, a basis is $\{c_1, c_2\}$. Applying Gram-Schmidt:

$$e_1 = \frac{c_1}{\|c_1\|} = \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \quad e_2 = \frac{c_2}{\|c_2\|} = \begin{bmatrix} 0 \\ -1 \end{bmatrix}.$$

$$C(A) = \text{span} \left\{ \begin{bmatrix} 1 \\ 0 \end{bmatrix}, \begin{bmatrix} 0 \\ -1 \end{bmatrix} \right\}$$

Row space $C(A^T)$

Rows of A :

$$r_1 = [1, 0, 3], \quad r_2 = [0, -1, 2].$$

They are independent. Applying Gram-Schmidt:

$$e_1 = \frac{r_1}{\|r_1\|} = \frac{1}{\sqrt{10}} \begin{bmatrix} 1 \\ 0 \\ 3 \end{bmatrix}, \quad u_2 = r_2 - \frac{\langle r_2, e_1 \rangle}{\|e_1\|^2} e_1 = r_2 - \frac{6}{1} e_1 = \begin{bmatrix} -6/\sqrt{10} \\ -1 \\ -16/\sqrt{10} \end{bmatrix},$$
$$e_2 = \frac{u_2}{\|u_2\|} = \frac{1}{\sqrt{(-6/\sqrt{10})^2 + (-1)^2 + (-16/\sqrt{10})^2}} \begin{bmatrix} -6/\sqrt{10} \\ -1 \\ -16/\sqrt{10} \end{bmatrix} = \frac{1}{\sqrt{38.2}} \begin{bmatrix} -6/\sqrt{10} \\ -1 \\ -16/\sqrt{10} \end{bmatrix}.$$

$$C(A^T) = \text{span}\{e_1, e_2\}$$

Nullspace $N(A)$

Solve $Ax = 0$:

$$x_1 + 3x_3 = 0 \implies x_1 = -3x_3, \quad -x_2 + 2x_3 = 0 \implies x_2 = 2x_3.$$

Let $x_3 = t$:

$$x = t \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}, \quad e = \frac{1}{\sqrt{14}} \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix}.$$

$$N(A) = \text{span} \left\{ \frac{1}{\sqrt{14}} \begin{bmatrix} -3 \\ 2 \\ 1 \end{bmatrix} \right\}$$

Left nullspace $N(A^T)$

Solve $A^T y = 0$:

$$\begin{bmatrix} 1 & 0 \\ 0 & -1 \\ 3 & 2 \end{bmatrix} \begin{bmatrix} y_1 \\ y_2 \end{bmatrix} = 0 \implies y_1 = 0, y_2 = 0.$$

$$\boxed{N(A^T) = \{0\}}$$

7. Consider the least square problem,

$$\min_x \frac{1}{2} \|Ax - b\|^2.$$

For the following data points:

$$\{(-1, 1), (0, 0), (1, 2), (2, 2)\},$$

- Find the closest straight line to them.
- Compute the error.
- Check that the error vector is perpendicular to columns of the matrix A .

Answer.

We are looking for the closest straight line of the form $y = mx + c$ to these points. The vector $\begin{bmatrix} m \\ c \end{bmatrix}$ represents the slope and the y -intercept of the line. The relationship between the data points (x_i, y_i) and the equation $y = mx + c$ can be written as:

$$A \begin{bmatrix} m \\ c \end{bmatrix} \approx b,$$

where A is the matrix formed from the x -coordinates and a column of ones, and b is the vector of y -coordinates.

For the given data points:

$$\{(-1, 1), (0, 0), (1, 2), (2, 2)\},$$

the x -coordinates are $-1, 0, 1, 2$ and the y -coordinates are $1, 0, 2, 2$. Hence:

$$A = \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix}, \quad b = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 2 \end{bmatrix}.$$

The least squares solution satisfies the normal equation:

$$A^T A \begin{bmatrix} m \\ c \end{bmatrix} = A^T b.$$

$$A^T A = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} = \begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix}, \quad A^T b = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix}.$$

$$\begin{bmatrix} 6 & 2 \\ 2 & 4 \end{bmatrix} \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 5 \\ 5 \end{bmatrix} \implies \begin{bmatrix} m \\ c \end{bmatrix} = \frac{1}{20} \begin{bmatrix} 4 & -2 \\ -2 & 6 \end{bmatrix} \begin{bmatrix} 5 \\ 5 \end{bmatrix} = \begin{bmatrix} 0.5 \\ 1 \end{bmatrix}.$$

Thus, the closest straight line is:

$$y = 0.5x + 1.$$

The error vector is:

$$e = b - A \begin{bmatrix} m \\ c \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} -1 & 1 \\ 0 & 1 \\ 1 & 1 \\ 2 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ 1 \end{bmatrix} = \begin{bmatrix} 1 \\ 0 \\ 2 \\ 2 \end{bmatrix} - \begin{bmatrix} 0.5 \\ 1 \\ 1.5 \\ 2 \end{bmatrix} = \begin{bmatrix} 0.5 \\ -1 \\ 0.5 \\ 0 \end{bmatrix}.$$

$$A^T e = \begin{bmatrix} -1 & 0 & 1 & 2 \\ 1 & 1 & 1 & 1 \end{bmatrix} \begin{bmatrix} 0.5 \\ -1 \\ 0.5 \\ 0 \end{bmatrix} = \begin{bmatrix} 0 \\ 0 \end{bmatrix}.$$

Thus, $A^T e = 0$, confirming that the error vector is perpendicular to the columns of A .
